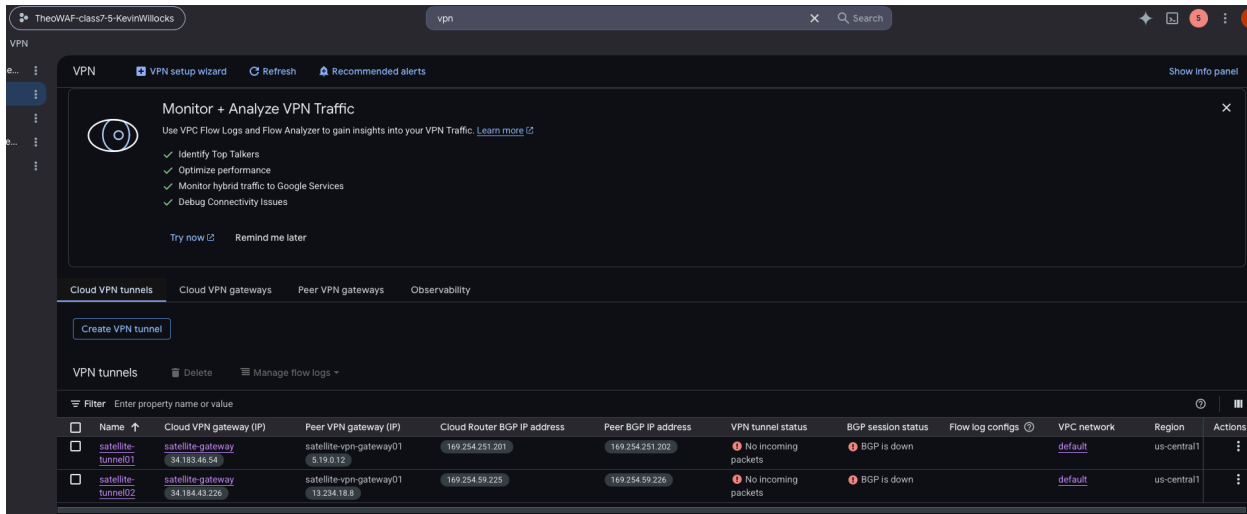


IPSec VPN + BGP Assessment

- 1) According to RFC 4301, what is the primary purpose of IPSec?
- B. To secure IP communications through authentication and encryption



The screenshot shows the AWS VPN console interface. At the top, there's a navigation bar with 'VPN' and a search bar. Below that, a 'Monitor + Analyze VPN Traffic' panel offers options to 'Try now' or 'Remind me later'. The main section is titled 'VPN tunnels' and includes a 'Create VPN tunnel' button. Below this is a table listing two VPN tunnels, 'satellite-tunnel01' and 'satellite-tunnel02'. Each tunnel is associated with a 'satellite-gateway' and a 'satellite-vpn-gateway01'. The table also shows BGP IP addresses and the status of the VPN tunnels and BGP sessions. Both tunnels show 'No incoming packets' and 'BGP is down'.

	Name	Cloud VPN gateway (IP)	Peer VPN gateway (IP)	Cloud Router BGP IP address	Peer BGP IP address	VPN tunnel status	BGP session status	Flow log configs	VPC network	Region	Actions
☐	satellite-tunnel01	satellite-gateway 34.183.46.54	satellite-vpn-gateway01 5.19.0.12	169.254.251.201	169.254.251.202	No incoming packets	BGP is down		default	us-central1	
☐	satellite-tunnel02	satellite-gateway 34.184.43.226	satellite-vpn-gateway01 13.234.18.8	169.254.59.226	169.254.59.226	No incoming packets	BGP is down		default	us-central1	

- 2) According to RFC 7296, which protocol version is used for modern IKE negotiation?

C. IKEv2

←

VPN tunnel details

🗑️ Delete

🕒 History

IP address

5.19.0.12

Cloud VPN gateway

Name

[satellite-gateway](#) (high-availability gateway)

VPC network

[default](#)

Region

us-central1

IP version

IPv4

IP stack type

IPv4 (single-stack)

IP address

Interface: 0 34.183.46.54

VPN Gateway logs

[View](#)

Routing and security

Routing type

Dynamic (BGP)

Cloud router

[satellite-router](#)

IKE version

IKEv2

IKE Phase 1

Encryption

Default

Integrity

Default

PRF

Default

Diffie-Hellman (DH)

Default

IKE Phase 2

Encryption

Default

Integrity

Default

PFS

Default

3) Which UDP port is primarily used for IKE / ISAKMP negotiations?

B. UDP 500

4) According to RFC 3948, which UDP port is commonly used for NAT Traversal (NAT-T)

C.UDP 4500

5) What is the primary purpose of a Pre-Shared Key (PSK) in IPSec?

B.To authenticate VPN peers

Google Cloud TheoWAF-class7-5-KevinWillocks vpn

Network Connectivity / VPN / Create a VPN

Network Connectivity Center

- VPN
- Interconnect
- Partner Cross-Cloud Inte...
- Cloud Router

Create a VPN

- Create Cloud HA VPN gateway
- Add VPN tunnels
- Configure BGP sessions
- Summary and reminder

VPN or Interconnect, [learn how global dynamic routing may affect you](#).

VPN tunnels

Expand each item to fill in the details of created VPN tunnels

Edit VPN tunnel

Associated Cloud VPN gateway interface
0 : 34.183.40.115

Associated peer VPN gateway interface
0 : 34.183.46.54

Name *
hyrule-tunnel01

Lowercase letters, numbers, hyphens allowed

Description

IKE version
IKEv2

IKE pre-shared key *
p649hfME7PJJCyHI48YmoWaU4Th91pa5 [Generate](#)

Enter your own key or generate one automatically

Make sure you record the pre-shared key in a secure location. The key can't be retrieved after this form is closed. [Learn more](#)

VPN cipher selection

If default option is selected or none of the ciphers are selected, it will automatically have default value [Learn more](#)

6) Which IPSec component is responsible for encrypting data traffic?

B. ESP

Google Cloud TheoWAF-class7-5-KevinWillocks vpn

Network Connectivity / VPN / VPN tunnel: satellite-tunnel02

Network Connectivity Ce... VPN Interconnect Partner Cross-Cloud Inte... Cloud Router

VPN tunnel details Delete History

Remote peer gateway

Name satellite-vpn-gateway01
IP address 13.234.18.8

Cloud VPN gateway

Name [satellite-gateway](#) (high-availability gateway)
VPC network [default](#)
Region us-central1
IP version IPv4
IP stack type IPv4 (single-stack)
IP address Interface: 1 34.184.43.226
VPN Gateway logs [View](#)

Routing and security

Routing type Dynamic (BGP)
Cloud router [satellite-router](#)
IKE version IKEV2

IKE Phase 1

Encryption	Default
Integrity	Default
PRF	Default
Diffie-Hellman (DH)	Default

IKE Phase 2

Encryption	Default
Integrity	Default
PFS	Default

7. What is the purpose of the Cloud Router in GCP?

C. Exchange BGP routing information

8. Which RFC defines the Encapsulating Security Payload (ESP)?

A. RFC 4303

Google Cloud

TheoWAF-class7-5-KevinWillocks

vpn

X

Q Search

17

Network Connectivity / Routers / Router: satellite-router

Network Connectivity Ce...

VPN

Interconnect

Partner Cross-Cloud Inte...

Cloud Router

Router details

Edit

Delete

Histor

satellite-router

Overview

Advertised and learned routes

Best routes

Router overview

Network

default

Region

us-central1

Interconnect encryption

Unencrypted

Cloud Router ASN

65501

BGP keepalive interval

60 seconds

BGP identifier

169.254.9.16/30

Tags

—

Advertised route configuration

BGP sessions will advertise these routes if no other configuration is specified

Advertisement mode

Default

BGP sessions

VPNSec BGP sessions

	Name ↑	Peer ASN	Type	Multiprotocol BGP	BGP Route policies	Cloud Router BGP IP
	domino-bgp02	65503	IPv4	Disabled IPv6		169.254.55.49
	satellite-bgp	65502	IPv4	Disabled IPv6		169.254.251.201
	satellite-bgp02	65503	IPv4	Disabled IPv6		169.254.59.225

9. Which protocol and port are used by BGP?

B. TCP 179



satellite-router

- Overview
- Advertised and learned routes
- Best routes

Router overview

Network	default
Region	us-central1
Interconnect encryption	Unencrypted
Cloud Router ASN	65501
BGP keepalive interval	60 seconds
BGP identifier	169.254.9.16/30
Tags	— 

Advertised route configuration

BGP sessions will advertise these routes if no other configuration is specified

Advertisement mode
Default

BGP sessions

VPNSec BGP sessions

oute policies	Cloud Router BGP IP	BGP peer IP	VPN gateway	VPN tunnel	Advertised route pric
	169.254.55.49	169.254.55.50		domino-tunnel02	
	169.254.251.201	169.254.251.202	satellite-gateway	satellite-tunnel01	
	169.254.59.225	169.254.59.226	satellite-gateway	satellite-tunnel02	

10. What is the purpose of the 169.254.x.x addresses used in HA VPN BGP sessions?

C. Link-local BGP peer communication

11. According to RFC 4271, what is the purpose of BGP?

B. Dynamically exchange routing information

← BGP session details Edit Delete

satellite-bgp

Status Down

Cloud Router [satellite-router](#)

Peer ASN 65502

Type IPv4

Multiprotocol BGP Disabled IPv6

Cloud Router BGP IP address 169.254.251.201

BGP peer IP address 169.254.251.202

VPN gateway [satellite-gateway](#)

VPN tunnel [satellite-tunnel01](#)

Peer connection Enabled

MD5 Authentication Disabled

Advertised route configuration

Advertisement mode

Cloud Router satellite-router settings

Advertised routes

Destination IP ranges ↑	Priority	Next hop
10.128.0.0/20	100	169.254.251.201

BGP Route policies

Filter Enter property name or value ⓘ

Name	Type	Term count	Position
No rows to display			

IP address

5.19.0.12

Cloud VPN gateway

Name

[satellite-gateway](#) (high-availability gateway)

VPC network

[default](#)

Region

us-central1

IP version

IPv4

IP stack type

IPv4 (single-stack)

IP address

Interface: 0 34.183.46.54

VPN Gateway logs

[View](#)

Routing and security

Routing type

Dynamic (BGP)

Cloud router

[satellite-router](#)

IKE version

IKEv2

IKE Phase 1

Encryption

Default

Integrity

Default

PRF

Default

Diffie-Hellman (DH)

Default

IKE Phase 2

Encryption

Default

Integrity

Default

PFS

Default

12. What is the most common cause of Phase 1 IPSec failures?

C. PSK mismatch

Filter
Enter property name or value

☐	Name ↑	Cloud VPN gateway (IP)	Peer VPN gateway (IP)	Cloud Router BGP IP address	Peer BGP IP address	VPN tunnel status	BGP session stat
☐	satellite-tunnel01	satellite-gateway 34.183.46.54	satellite-vpn-gateway01 5.19.0.12	169.254.251.201	169.254.251.202	🚫 No incoming packets	🚫 BGP is down
☐	satellite-tunnel02	satellite-gateway 34.184.43.226	satellite-vpn-gateway01 13.234.18.8	169.254.59.225	169.254.59.226	🚫 No incoming packets	🚫 BGP is down

Cloud VPN gateway

Name	satellite-gateway (high-availability gateway)
VPC network	default
Region	us-central1
IP version	IPv4
IP stack type	IPv4 (single-stack)
IP address	Interface: 0 34.183.46.54
VPN Gateway logs	View ↗

— — — — —

13. Which of the following is typically configured on both VPN peers?

C. Matching encryption settings

14. Which GCP component creates the public IP addresses used by the VPN tunnels?

B. HA VPN Gateway

15. What BGP session state indicates successful route exchange?

D. Established

16.Which IPSec protocol uses IP Protocol 50?

B. ESP

17.Why do companies commonly deploy dual HA VPN tunnels?

B. For redundancy and failover

18. What is the primary purpose of NAT Traversal (NAT-T)?

C.Allow IPSec traffic through NAT devices

19. Which of the following best describes a Security Association (SA)?

B. A set of agreed IPsec security parameters

20. What is the correct order of IPsec and BGP establishment?

B. IKE Phase 1 → IPsec Phase 2 → BGP